From acceleration to velocity

Constant acceleration produces equal velocity changes over equal time intervals.

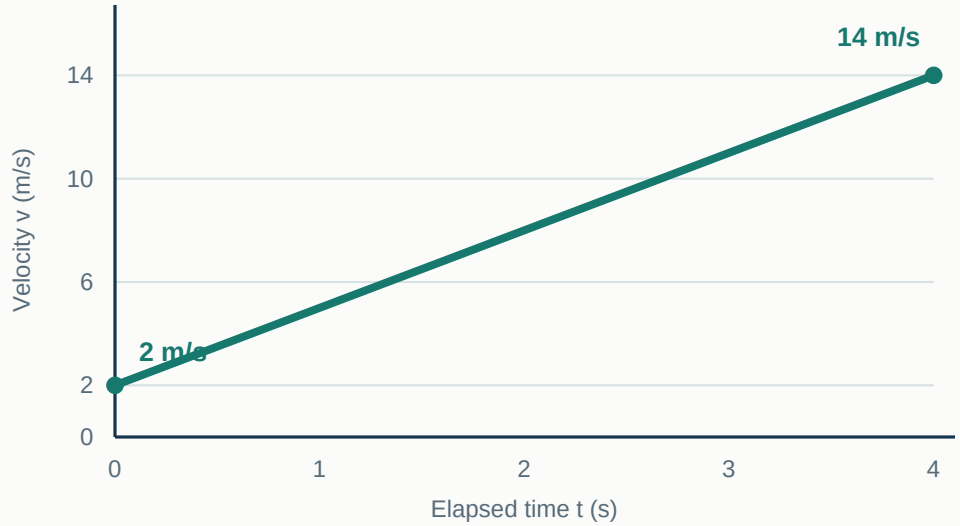
Initial velocity: 2 m/s

Constant acceleration: 3 m/s per second

Observation interval: 0 to 4 s

$$v(t) = v_0 + a\,t$$

v0 = initial velocity a = acceleration t = time



The line's slope is acceleration; its starting height is initial velocity.

Displacement is the area under velocity

For this motion, the velocity-time region is a trapezoid.
Its area equals the displacement over the interval.

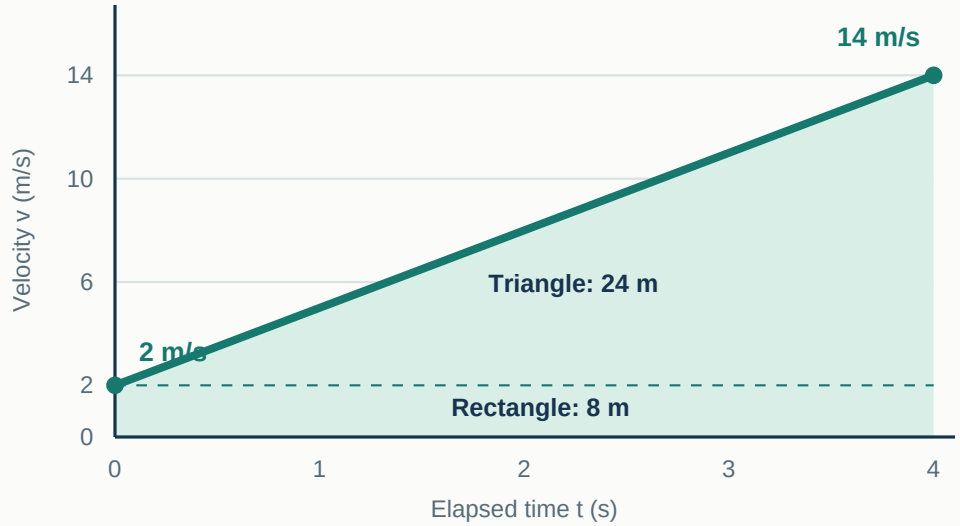
Average velocity is the mean of the endpoints.

Multiply average velocity by elapsed time.

$$\bar{v} = \frac{v_0 + v(t)}{2}$$

$$\Delta x = \bar{v}\, t$$

v-bar = average velocity delta x = displacement



The graph converts a velocity calculation into a position change.

Build the constant-acceleration equation

Substitute the velocity expression into the area relation.

Distribute the elapsed-time factor over both terms.

$$\Delta x = \frac{v_0 + (v_0 + a\,t)}{2}\,t$$

$$\Delta x = v_0\,t + \frac{1}{2}a\,t^2$$

Existing motion contributes the rectangular area.

Acceleration contributes the additional triangular area.

delta x = position change v0 = initial velocity a = acceleration t = elapsed time

Every term has units of length; this model requires constant acceleration.